

core rules

$$\mathcal{L}\{af + bg\} = aF + bG$$

$$\mathcal{L}\{f'\} = sF - f(0^+)$$

$$\mathcal{L}\{\int_0^t f(\tau) d\tau\} = F(s)/s$$

useful limits and shifts

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$$

$$f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$